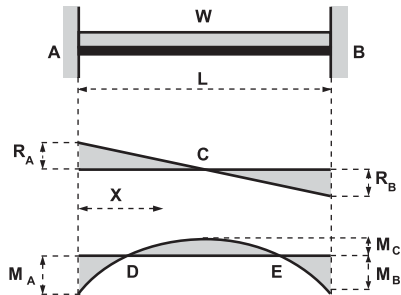


Beam fixed at both ends

Uniform load on full span

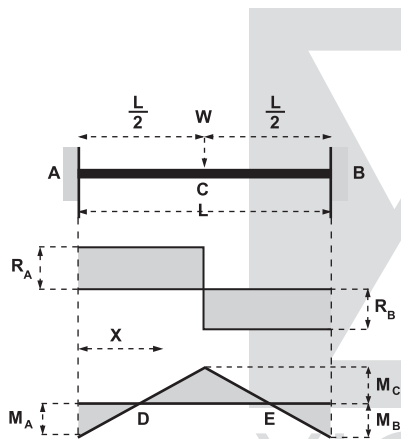


$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total Uniform load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 M_A = M_B &= -\frac{WL}{12} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_C &= \frac{WL}{24} \\ \delta_{\max} &= \frac{WL^3}{384EI} \end{aligned} \right. \\
 \text{at X from A} &\left\{ \begin{aligned} M_X &= \frac{W}{12L}(L^2 - 6LX + 6X^2) \\ \delta_X &= \frac{WX^2}{24EI}(L-X)^2 \\ i_X &= \frac{WX}{12EI}(L^2 - 3LX + 2X^2) \end{aligned} \right.
 \end{aligned}$$

at 0.211L from either end $M_D = M_E = 0$

Beam fixed at both ends

Point load at mid-span

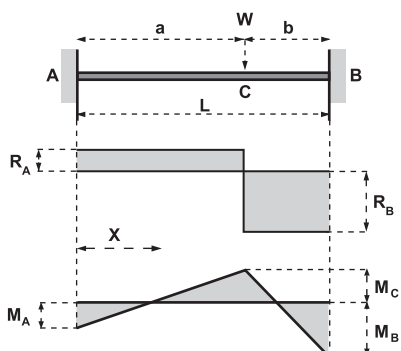


$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 M_A = M_B &= -\frac{WL}{8} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_C &= \frac{WL}{8} \\ \delta_{\max} &= \frac{WL^3}{192EI} \end{aligned} \right. \\
 \text{at X from A between A and C} &\left\{ \begin{aligned} M_X &= \frac{W}{8}(4X - L) \\ \delta_X &= \frac{WX^2}{48EI}(3L - 4X) \\ i_X &= \frac{WX}{8EI}(L - 2X) \end{aligned} \right.
 \end{aligned}$$

at 0.25L from either end $M_D = M_E = 0$

Beam fixed at both ends

Point load at any position



$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A &= \frac{Wb^2(L+2a)}{L^3} & R_B &= \frac{Wa^2(L+2b)}{L^3} \\
 M_A &= -\frac{Wab^2}{L^2} & M_B &= -\frac{Wa^2b}{L^2} \\
 \text{at C, under load, } M_C &= \frac{2Wa^2b^2}{L^3} \\
 \text{at X from A between A and C} &\left\{ \begin{aligned} M_X &= \frac{Wab^2}{L^2} \frac{Wb^2(L+2a)X}{L^3} + \\ \delta_X &= \frac{Wb^2X^2[3La - (L+2a)X]}{6EIL^3} \\ i_X &= \frac{Wb^2X[2La - (L+2a)X]}{2EIL^3} \\ \delta_{\max} &= \frac{2Wa^3b^2}{3EI(L+2a)^2} \end{aligned} \right.
 \end{aligned}$$

when $a > b$ the maximum deflection is at $X = \frac{2La}{L+2a}$